

Accuracy–Efficiency Trade-Off Analysis of Deep Learning Models for Breast Cancer Diagnosis using Ultrasound Images

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Abstract—This paper presents a comparative analysis of four deep learning architectures on an ultrasound image dataset of 2104 training and 529 test images of human breast, differentiating between three classes (benign, malignant, or normal). These models are evaluated on multiple parameters -accuracy loss, Cohen’s Kappa coefficient, Area Under Curve (AUC), and computational efficiency. The results show InceptionV3 achieves the highest overall accuracy of 94.14% with a Cohen’s Kappa score of 0.9121, followed by DenseNet121, which provided competitive performance with 93.01% accuracy and the lowest test loss of 0.2024. VGG16 delivered an accuracy of 92.06%, whereas MobileNetV2 showed a lower accuracy of 86.96%. Robust comparisons between models are ensured using statistical significance testing and highly sophisticated evaluation metrics. By tackling issues with model selection, accuracy over computational efficiency trade-offs, and deployment considerations, this study offers helpful insights for well-informed choices in deep learning applications.

Index Terms—Deep Learning, Ultrasound Image Classification, InceptionV3, DenseNet121, VGG16, MobileNetV2

I. INTRODUCTION

Deep learning technologies, particularly Convolutional Neural Networks (CNNs), have made the computer vision area progress faster than it ever has [1]. These architectures have revolutionized image classification, achieving human-level or even superior performance across many areas such as medical imaging, self-driving cars, facial recognition, and content moderation systems [2], [3]. CNN models function well because they can automatically learn hierarchically structured features representations from untaught batches of raw image pixel values, thus eliminating the need for feature engineering

present in previous generations of non-deep learning methods for driving performance [4], [5].

CNN architecture has progressed through a succession of breakthroughs ImageNet Large Scale Visual Recognition Challenge (ILSVRC), then the deeper VGG networks, the introduction of residual connections in ResNet, and Finally, focus on an efficiency-optimized architecture like MobileNet [6]–[8]. Each architecture serves to solve certain limitations of its predecessor, while also introducing new design ideas for future architectures.

II. RELATED WORK

With each successive generation overcoming challenges, Convolutional neural networks (CNNs) have come a long way from AlexNet. VGG networks demonstrated that deeper models with smaller filters are better than shallow networks with big filters, and ResNet introduced skip connections to counter vanishing gradients in very deep models. DenseNet optimized parameters further by fac. Most of the studies consider single architectures or binary classification problems. Comparison on equal grounds is often lacking. For instance, Bianco et al. [9] focused on ImageNet performance rather than domain-specific, facilitating dense connectivity and feature reuse [10]–[12]. CNNs have also achieved expert-level performance in medical imaging, such as dermatologist-level skin cancer classification Esteva et al. [13], and lung cancer diagnosis with chest X-rays [14]. The majority of studies take one architecture into account or perform binary classification

tasks. Comparative analyses are normally plagued by uneven standards. For example, Bianco et al. [9] concentrated on ImageNet performance versus domain-specific outcomes, and Canziani et al. [15] compared computational efficiency but limited their effort to medical image work. This contribution provides an extensive, multi-faceted Examination of CNNs in medical image classification, including statistical significance testing and deployment considerations, has largely avoided in the literature.

III. METHODOLOGY

The experiment utilized a set of 2,633 breast ultrasound images labelled as benign, malignant, and normal. A total of 2,104 images were assigned to training and 529 to testing [16]. The dataset was moderately imbalanced, with normal cases representing 44.3%, benign and malignant cases representing 28.2% and 27.5%, respectively. Such an imbalance necessitated the application of stable evaluation metrics to guarantee solid judgment across all classes.

This study employs a publicly available breast ultrasound imaging dataset sourced from Kaggle, comprising three classes of images: benign, malignant, and normal [17]. Figure 1 illustrates the dataset contained both original ultrasound scans and their corresponding segmentation masks. It enables both effective supervised training and evaluation of deep learning models for tumor classification [18]. This dataset is very suitable benchmark for evaluation of performance of deep learning based diagnostic model due to it's diverse image characteristics including noise patterns, clinical artifacts and varying lesion appearances.

The experiments were carried out in the free tier environment of Google Colab with its hosted GPU capabilities. The environment was set with Python 3.8, TensorFlow 2.8.0, and Keras 2.8.0, along with CUDA and cuDNN for hardware support. During the experiment, all the images were resized to 224×224 pixels for uniform input and normalized with ImageNet mean and standard deviation. An identical data augmentation pipeline was applied during training to ensure a equitable and congruous comparison across all deep learning models. The augmentation strategy comprised random rotation of $\pm 20^\circ$, width and height translation of 0.2, random zoom of 0.2, and shuffling with rotation-based augmentation enabled for all models [19]–[21]. A 20% validation split was reserved for uniform model tuning.

All four models were selected based on their suitability for image based classification. VGG16 serves a widely adopted baseline and is extensively used in medical imaging applications [37]. DenseNet121 is well suited for smaller datasets due to it's feature reuse capability through dense connectivity [38]. MobileNetV2 is lightweight and efficiency focused and suitable for resource-constrained environments [39]. Finally InceptionV3 performs multiscale feature extraction through parallel convolutional operations [40].

All four CNN architecture-VGG16, DenseNet121, MobileNetV2, and InceptionV3 was pretrained on ImageNet initial weights through transfer learning [6], [22], [23]. The

training was done in two stages. Initially, to learn the dataset effectively in a brief amount of time, the custom classification heads were trained with a higher learning rate in isolation. Later, fine-tuning was done on the entire network with a lower learning rate of $1e-4$; a detailed training description is given in Table III. The Adam optimizer has a batch size 32 and categorical cross-entropy loss, early stopping with weight restoration to prevent overfitting was used [24]. Performance measures included precision, recall, accuracy, Cohen's Kappa, F1-score, and AUC, as mentioned in Table I, and are complemented by computational measures such as training time and parameter count [25]–[28].

TABLE I
EVALUATION METRICS USED IN THIS STUDY

Criteria	Formula
Recall (R_t)	$R_t = \frac{TP}{TP + FN}$
Precision (P_t)	$P_t = \frac{TP}{TP + FP}$
F1 Score (F_{1t})	$F_{1t} = \frac{2 \times R_t \times P_t}{R_t + P_t}$
Classification Accuracy (C_{At})	$C_{At} = \frac{TP + TN}{TP + FP + TN + FN}$
Cohen's Kappa Score (K_t)	$K_t = \frac{2 \times (TP \times TN - FP \times FN)}{(TP + FP)(FP + TN) + (TP + FN)(TN + FN)}$

Note: TP = True Positives, TN = True Negatives, FP = False Positives, FN = False Negatives.

IV. ARCHITECTURE IMPLEMENTATION

In this study, four CNN architectures, VGG16, DenseNet121, MobileNetV2, and InceptionV3 were used for a three-class image classification task [22], [29]. All models used transfer learning with pre-trained ImageNet weights were adapted by removing the top layers (include_top=False) and adding a custom classification head. The majority of convolutional layers were frozen, and only the last $L \approx 30$ layers were fine-tuned [6], [23]. Images were normalized to the $[0, 1]$ range using rescaling. Training and validation were done using directory-based generators, with a batch size of $B = 20$ and $E = 10$ epochs. Optimization was done using the Adam optimizer with a learning rate of $\eta = 1 \times 10^{-4}$.

The classification task for all models was trained using sparse categorical cross-entropy:

$$\mathcal{L} = -\frac{1}{Z} \sum_{i=1}^Z \log p_\theta(y_i | x_i) \quad (1)$$

Equation (1) defines the terms, where Z is the number of training samples, x_i denotes the input image, y_i represents the true label, and $p_\theta(y_i | x_i)$ indicates the predicted softmax probability.

A. VGG16

The input size of the VGG16 was $224 \times 224 \times 3$. Thirteen convolutional layers with three-by-three kernels and three fully connected layers make up the base [30]. GlobalAveragePooling $\rightarrow$ Dense(1024, ReLU) $\rightarrow$ Dropout(0.5) $\rightarrow$ Dense(512, ReLU) $\rightarrow$ Dropout(0.3) $\rightarrow$ Dense(3, Softmax) was the custom head. There were about 138 million parameters in all.

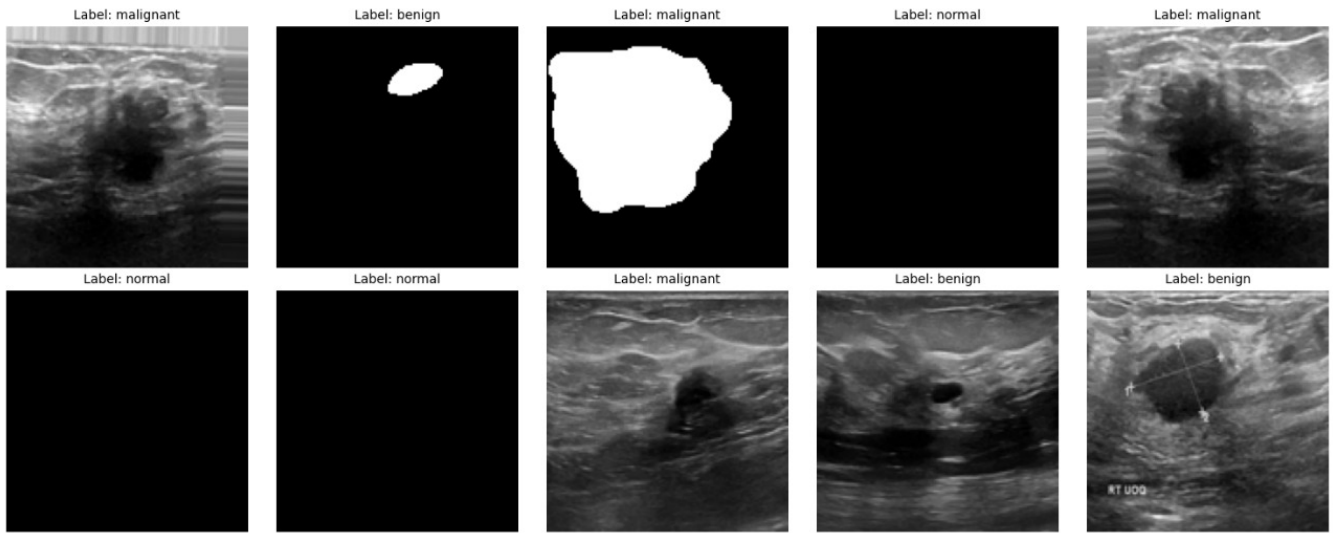


Fig. 1. Sample breast ultrasound input images representing benign, malignant, and normal classes used for model training and evaluation.

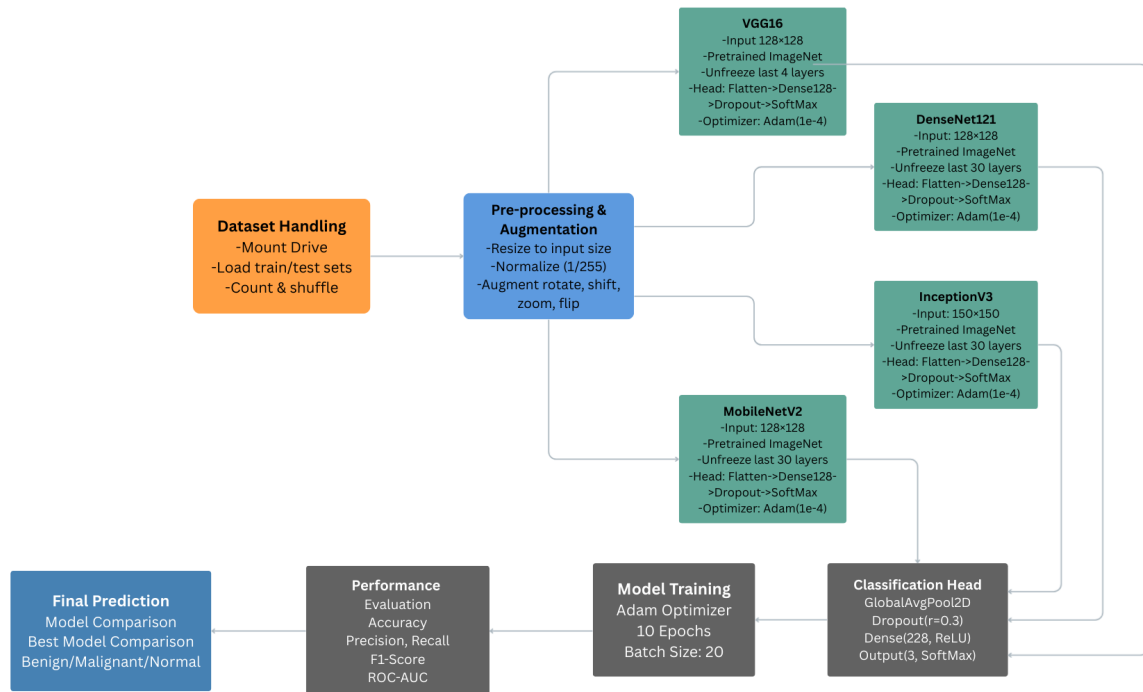


Fig. 2. Overall Architectural Diagram.

B. DenseNet121

Each layer in DenseNet121's dense connectivity architecture receives feature maps from every layer that came before it in the same dense block. Four dense blocks with 6, 12, 24, and 16 layers separated by transition layers follow the base's initial 7×7 convolution. The final block's output passes through GlobalAveragePooling $\rightarrow$ Dense(128, ReLU) $\rightarrow$ Dropout(0.3, 0.2) $\rightarrow$ Dense(3, Softmax). There are about 8 million parameters in the model.

C. MobileNetV2

Inverted residual blocks with depthwise separable convolutions are used in MobileNetV2. The size of the input was $128 \times 128 \times 3$. GlobalAveragePooling $\rightarrow$ Dense(128, ReLU) $\rightarrow$ Dropout(0.3, 0.2) $\rightarrow$ Dense(3, Softmax) is used to process the output of the final convolutional block. As a result, the model is lightweight and has only about 3.4 million parameters.

D. InceptionNetV3

Spatial correlations at various scales are captured by InceptionV3 using Inception modules. The size of the input was $150 \times 150 \times 3$. GlobalAveragePooling was used to pool the features after the convolutional base, and then Dense(128, ReLU) $\rightarrow$ Dropout(0.3, 0.2) $\rightarrow$ Dense(3, Softmax) was used. There are roughly 23.8 million parameters in all.

Figure 2 illustrates the complete workflow of a deep learning-based approach for breast cancer detection and classification firstly, data processing is done by resizing, normalizing, and augmenting the data to make the model more robust [31], [32]. All four pre-trained models are fine-tuned using a common classification head and later trained using the Adam optimizer. Model performance is evaluated through standard matrices this lead to selection of the best performing model for prediction across benign, malignant, and normal classes.

TABLE II
COMPARATIVE ANALYSIS OF DEEP LEARNING MODELS

Model	Input Size	Parameters	Base Details	Classification Head
VGG16	$224 \times 224 \times 3$	138M	13 Conv + 3 FC, 3×3 kernels	GAP $\rightarrow$ Dense(1024, ReLU) $\rightarrow$ Dropout(0.5) $\rightarrow$ Dense(512, ReLU) $\rightarrow$ Dropout(0.3) $\rightarrow$ Dense(3, Softmax)
DenseNet121	$128 \times 128 \times 3$	8M	4 Dense Blocks (6, 12, 24, 16), growth rate $k = 32$	GAP $\rightarrow$ Dense(128, ReLU) $\rightarrow$ Dropout(0.3, 0.2) $\rightarrow$ Dense(3, Softmax)
MobileNetV2	$128 \times 128 \times 3$	3.4M	Inverted residual blocks	GAP $\rightarrow$ Dense(128, ReLU) $\rightarrow$ Dropout(0.3, 0.2) $\rightarrow$ Dense(3, Softmax)
InceptionV3	$150 \times 150 \times 3$	23.8M	Inception modules A, B, C + reductions	GAP $\rightarrow$ Dense(128, ReLU) $\rightarrow$ Dropout(0.3, 0.2) $\rightarrow$ Dense(3, Softmax)

Note: GAP = Global Average Pooling, FC = Fully Connected, Conv = Convolutional.

Table II present a complete analysis of all four CNN models. It digest the input resolutions, parameter counts, and core

architectural characteristics. The models, includes layers such as GAP, Dense, ReLU, and Dropout. The base detail are very distinct and model specific [5]. Among all four models VGG16 has the highest number of parameter a lot greater than rest of the models. Substantial variation in model complexity and design philosophy across the chosen architectures is highlighted in this comparison.

V. RESULTS

The experiment was conducted using four deep learning models: InceptionV3, VGG16, DenseNet121, and MobileNetV2. The research outcomes have been compiled in the following tables and images, which present the evaluation metrics and performance comparisons of these models. The results indicate that InceptionV3 outperforms the other models, achieving the highest classification accuracy of 94.14% [33].

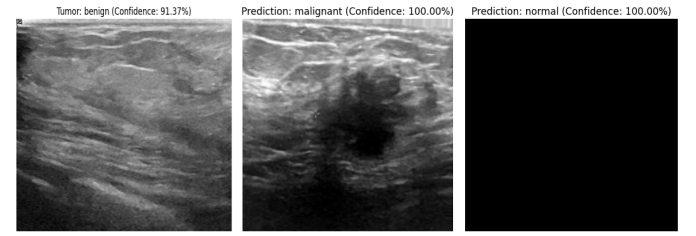


Fig. 3. Prediction outputs from the VGG16 model on sample breast ultrasound images showing benign, malignant, and normal tumor classifications with corresponding confidence scores.

Figure 3 illustrate the model's ability to distinguish between different breast tissue conditions with high accuracy. The three images represent detection of benign, malignant and no tumor detection with accuracy of 91.37%, 100% and 100% respectively. The drop in accuracy in benign is a result of class augmentation.

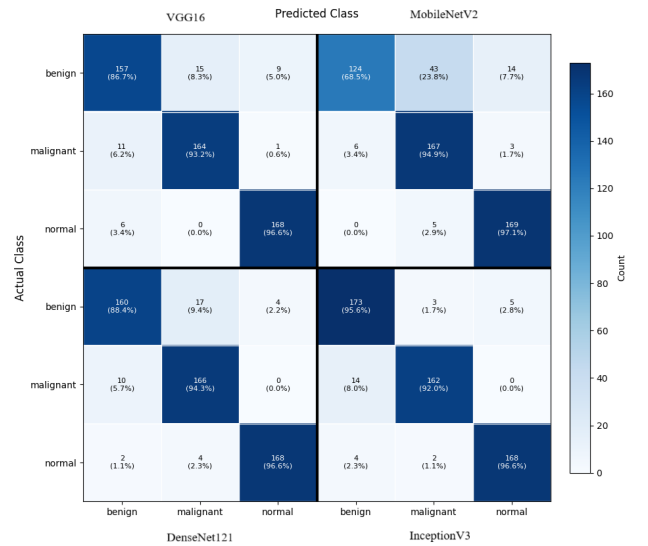


Fig. 4. Confusion Matrices for each model.

Based on the confusion matrix plotted in Figure 4, InceptionV3 demonstrates the highest overall recall rate across all three image classes (benign, malignant, and normal) showing better ability to distinguish between these three classes. Furthermore, DenseNet121 has demonstrated excellent recall among all categories, particularly showing strong performance in detecting malignant and normal. However, VGG16 delivers steady results but exhibits comparatively weaker differentiation between benign and malignant samples, whereas MobileNetV2, though computationally efficient, shows reduced sensitivity toward benign images, thereby lowering its overall classification accuracy.

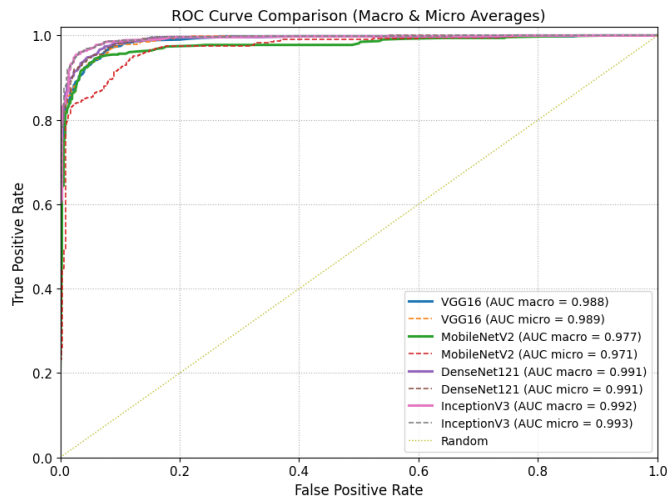


Fig. 5. AUC rankings for the models.

Figure 5 illustrates ROC curve comparison of all for CNN model using macro and micro averaged AUC values. All four models exhibit strong discriminative ability, with AUC values above 0.97. Among these models, InceptionV3 achieves the highest macro and micro AUC values, indicating its better overall predictive reliability. DenseNet121 and InceptionV3 demonstrate nearly perfect classification performance, reaching AUC score over 0.99.

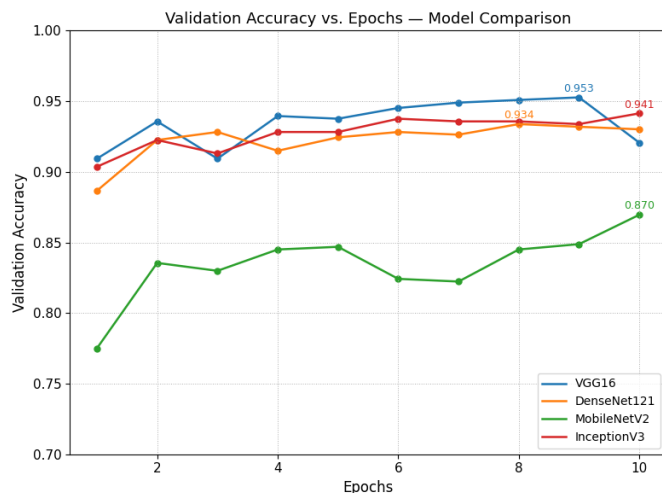


Fig. 6. Validation accuracy vs epoch graph.

Figure 6 illustrates the validation accuracy trend for all four CNN models over 10 epochs. A stable validation accuracy was shown by all four models which indicates that they generalize well. VGG16 shows the highest validation accuracy, with InceptionV3 following closely behind. MobileNetV2 has lower accuracy, designating a trade-off between validation performance and model complexity.

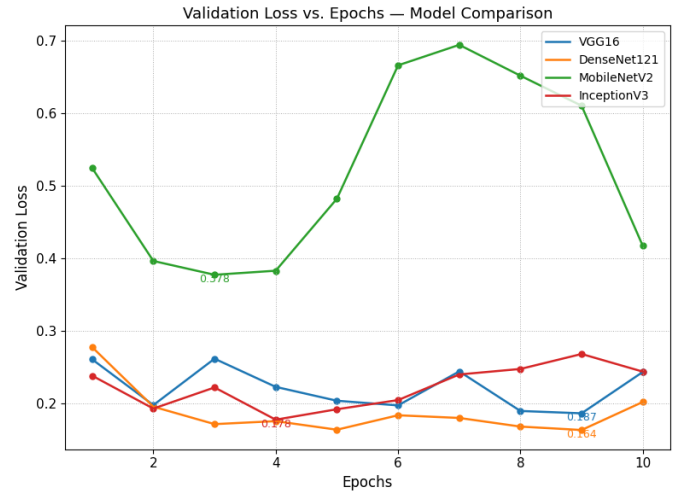


Fig. 7. Validation loss vs epoch graph.

Figure 7 illustrates the validation loss trend for all four CNN models over 10 epochs. A fluctuating, yet mostly stable, validation loss patterns were shown by all four models which indicates that they generalize well. DenseNet121 exhibits the lowest and most consistent loss, reflecting superior validation stability. In contrast, MobileNetV2 exhibits higher and more variable loss values, suggesting relatively weaker generalization performance.

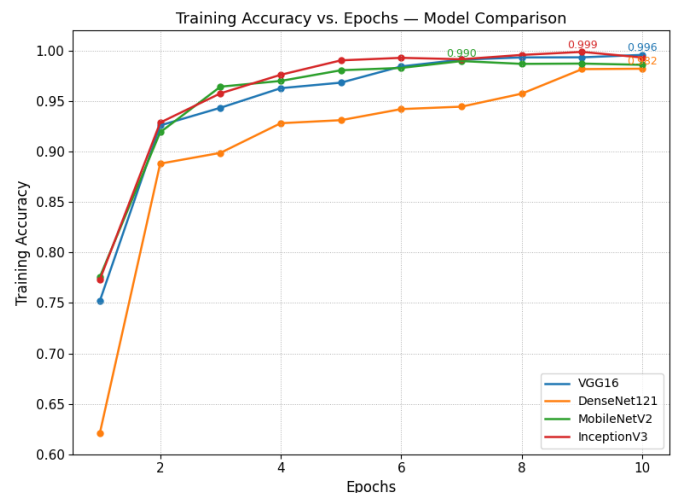


Fig. 8. Training accuracy vs epoch graph.

Figure 8 illustrates the training accuracy trend of all four CNN models over 10 epochs. A steady improvement in accuracy was shown by all four models. DenseNet121 coincides more slowly but eventually reaches stable performance after

several epochs. MobileNetV2 and InceptionV3 achieves the highest accuracy.

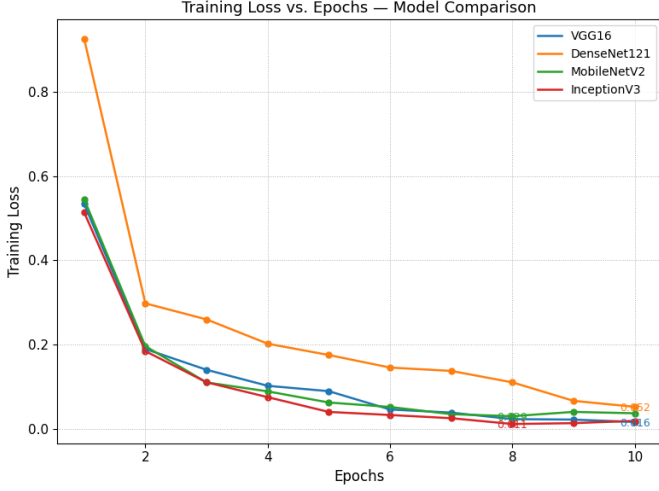


Fig. 9. Training loss vs epoch graph.

Figure 9 illustrates the loss progression trend for all CNN models over 10 epochs. A steady decline in accuracy was shown by all four models thus indicating effective learning and convergence. MobileNetV2 and InceptionV3 achieves the lowest loss value. DenseNet121 had relatively higher loss in early epochs but gradually stabilizes with later epochs.

TABLE III
PERFORMANCE AND COMPLEXITY COMPARISON OF CNN ARCHITECTURES

Techniques	Depth	No. of Parameters (M)	Computational Time (Avg. per Epoch, s)	Batch Size	Epochs
VGG16	16	~138M	557 s	32	10
DenseNet121	121	~8M	159 s	32	10
MobileNetV2	~53	~3.4M	120 s	32	10
InceptionV3	~48 modules (≈300 layers)	~23.9M	148 s	32	10

Table IV presents the performance of the four CNN models on the breast cancer classification dataset. It outlines the precision, recall, and F1-score for benign, malignant, and normal classes, along with overall accuracy, Cohen’s Kappa, macro-averaged AUC, p-value, and confidence interval (CI) for each model. A satisfactory performance was achieved by all four models, with InceptionV3 demonstrating the best overall performance. Notably, InceptionV3 significantly outperformed MobileNetV2 ($p < 0.05$), while its performance remained statistically comparable to VGG16 and DenseNet121.

A. Performance Metrics Analysis

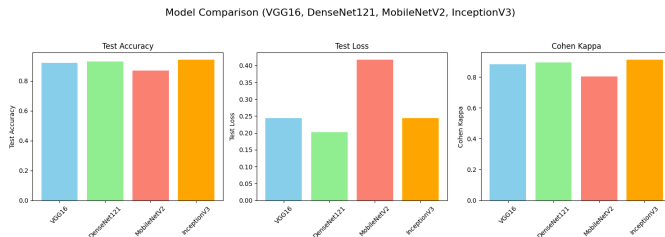


Fig. 10. Model Comparison (Test accuracy, Test loss, Cohen’s Kappa).

Figure 10 illustrates performance comparison of all four models in terms of test accuracy, test loss, and Cohen Kappa metrics. The overall results indicate that InceptionV3 maintains the most robust and well-organized performance across various evaluation metrics [41].

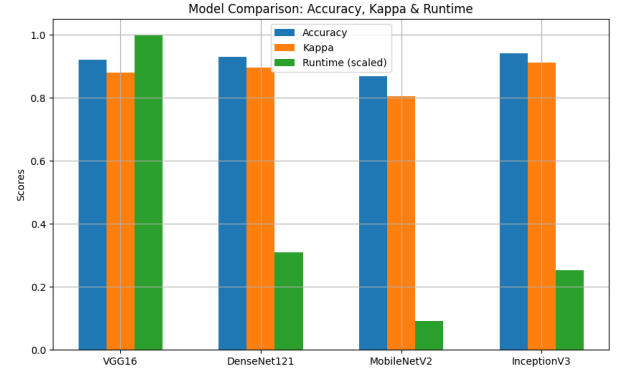


Fig. 11. Model Comparison (Accuracy, Cohen’s Kappa, Runtime).

Figure 11 illustrates the comparison of all four CNN models based on accuracy, Cohen’s Kappa, and normalized runtime metrics. MobileNetV2 has a slightly lower performance but offers the fastest inference time, which shows its efficiency. In case of clinical diagnosis tasks, accuracy can provide misleading performance estimates with class imbalance. Therefore, Cohen’s kappa is a more relevant metric, as it quantifies the level of agreement beyond chances.

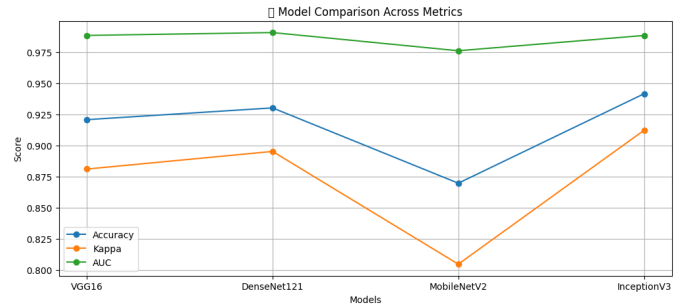


Fig. 12. Model Comparison (Accuracy, Cohen’s Kappa, AUC).

Figure 12 illustrates how all four CNN models performed on Accuracy, Cohen’s Kappa and AUC metrics. MobileNetV2 achieved the lowest in all three metrics highlighting its computational efficiency tradeoff. DenseNet121 scored a slightly higher accuracy and AUC which imply reliable and consistent predictions. InceptionV3 performs better than other models overall. Its higher accuracy and stronger agreement makes it the most dependable architecture among all four.

VI. DISCUSSION

The suggested enhancements highlight architectural advancements, training upgrades, assessment paradigms, and runtime plans for pushing forward medical image analysis

TABLE IV
RESULTS COMPARISON ON BREAST CANCER DATASET

Model	Class	Recall (R_t)	Precision (P_t)	F1 (F_{1t})	Accuracy (C_{At}) (%)	Kappa (K_t)	AUC	p-value	95% CI	Significance
VGG16	Benign	0.87	0.90	0.88	92.06	0.8809	0.987	0.0705	(−0.0017, 0.0432)	Non-significant
	Malignant	0.93	0.92	0.92						
	Normal	0.97	0.94	0.95						
DenseNet121	Benign	0.88	0.93	0.91	93.01	0.8951	0.990	0.1228	(−0.0051, 0.0428)	Non-significant
	Malignant	0.94	0.89	0.91						
	Normal	0.97	0.98	0.97						
MobileNetV2	Benign	0.69	0.95	0.80	86.96	0.8045	0.977	1.13×10^{-5}	(0.0359, 0.0922)	Significant
	Malignant	0.95	0.78	0.86						
	Normal	0.97	0.91	0.94						
InceptionV3	Benign	0.95	0.89	0.92	94.14	0.9121	0.987	–	–	–
	Malignant	0.92	0.95	0.94						
	Normal	0.95	0.98	0.97						

Note: Statistical significance is assessed using the Wilcoxon signed-rank test with InceptionV3 as the reference model.

[34], [35]. Ensemble techniques are a promising avenue through aggregating the complementary strengths of current architectures. An accuracy-centric ensemble of InceptionV3 and DenseNet121 would achieve maximum predictive potential, whereas an energy-balanced ensemble of MobileNetV2 and DenseNet121 would strive to balance accuracy against energy expenditure on portable platforms [7], [8]. Confidence-based weighted voting also reinforces ensemble credibility. Attention mechanisms can also be used to improve diagnostic accuracy. Spatial attention would allow the model to focus on the most informative areas of images, and channel attention highlights important feature maps. Self-attention also maintains long-distance spatial relationships and therefore augments feature representation. Neural Architecture Search (NAS) also allows for auto-optimization, optimizing for accuracy within computational budgets, optimizing for reduced resource usage with still-acceptable accuracy, or optimizing for multi-objective trade-offs between both [36].

Aside from architectural refinements, there are methodological refinements to be made for improvements of generalizability. Higher-level augmentation approaches—domain-specific, adversarial, or generative—augment the variability of data. Optimization techniques, adaptive learning rate, progressive training, and curriculum learning also aid convergence. Calibration and robustness are enhanced using regularization techniques such as knowledge distillation and dropout variants. Thirdly, deployment-focused approaches are also required. Distributed and federated training enable the development of scalable and privacy-robust models. Model size is minimized using distillation, quantization, and so on. Hardware-optimized customizations and dynamic inference and offline processing help enable efficient real-world deployment. The proposed classification framework employs a transfer learning-based InceptionV3 architecture for the prediction task, as detailed in Algorithm 1. The input images of size $150 \times 150 \times 3$ are preprocessed and passed through an ImageNet-initialized InceptionV3 base network. The final 30 layers of the model are selectively unfrozen for fine-tuning. A custom classifi-

cation head consisting of two ReLU-activated dense layers, Global Average Pooling (GAP), and dropout regularization is appended to enhance feature discrimination. The model is trained using the Adam optimizer with a learning rate of 1×10^{-4} for 10 epochs. This configuration ensures efficient feature extraction while improving generalization performance on the target dataset.

Algorithm 1: Proposed Transfer Learning-Based InceptionV3 Model

- 1: **Input:** Images of size $150 \times 150 \times 3$
- 2: **Output:** Predicted class labels (Class 1, Class 2, Class 3)
- 3: Set learning rate $\eta = 1 \times 10^{-4}$, batch size $B = 20$, epochs $E = 10$, dropout probabilities $p_1 = 0.3$, $p_2 = 0.2$
- 4: Load dataset and rescale pixel values to $[0, 1]$
- 5: Initialize InceptionV3 base model with ImageNet weights (`include_top = False`)
- 6: Freeze all layers of the base model
- 7: Unfreeze last $L \approx 30$ layers for fine-tuning
- 8: Construct classification head:
 - a) GlobalAveragePooling2D
 - b) Dense layer (128 neurons, ReLU)
 - c) Dropout layer ($p = 0.3$)
 - d) Dense layer (128 neurons, ReLU)
 - e) Dropout layer ($p = 0.2$)
 - f) Dense layer ($C = 3$, Softmax)
- 9: Compile model using Adam optimizer with learning rate η
- 10: Train model using training generator:
 - a) Steps per epoch = length of training generator
 - b) Validation data = test generator
 - c) Validation steps = length of test generator
- 11: Evaluate model performance on test dataset
- 12: Return predicted class labels

VII. CONCLUSION

This work conducted an exhaustive validation of four leading convolutional neural network (CNN) architectures under the multi-class image classification scenario with medical images. It exposes a definitive hierarchy of performance, wherein InceptionV3 delivered the highest accuracy (94.14%) and Cohen's Kappa (0.9121), DenseNet121 showed excellent generalisability and superb parameter parsimony, VGG16 was a solid baseline, and MobileNetV2 delivered computational efficiency for low-resource settings. These findings expose important architectural trade-offs between accuracy, parameter parsimony, and computational overhead, thus serving as a useful guide to model selection driven by deployment constraints. This study presents an integrated assessment framework and selection guidelines for architecture in clinical and real-world settings. The research highlights the congruence of architectural advantages with application-specific needs, ranging from critical diagnosis to point-of-care utilisation. This work lays good groundwork for further exploration, even after being limited to a single dataset, a single spectrum of architectural options and dependence on transfer learning. Areas for improvement are validation on a range of datasets, exploration of newer architectures, including vision transformers, a thorough investigation of the ensemble schemes, and domain adaptation to facilitate cross-modal generalisability. Overall, this work provides a systematic baseline for CNN-based classification of medical images and provides both theoretical understanding and practical suggestions to help researchers, practitioners, and designers of systems in designing accurate, efficient, and clinically feasible solutions for deep learning. Its conclusions ultimately affirm that the selection of architecture should be dictated not just by a consideration of accuracy, but by a consideration of the practical limitations and clinical requirements of the target environment for final use.

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